Assignment - 5

Installation of Hive on single node cluster of hadoop.

Steps to install Hive on single node cluster of hadoop -

1. Start the Hadoop cluster

1.1. start-dfs.sh && start-yarn.sh

1.2. Jps

```
Starting namenodes on [localhost]
Starting datanodes
localhost: ERROR: Cannot set priority of datanode process 18906
Starting secondary namenodes [localhost]
localhost: ERROR: Cannot set priority of secondarynamenode process 19119
Starting resourcemanager
Starting nodemanagers
localhost: ERROR: Cannot set priority of nodemanager process 19709
```

```
18720 NameNode
21986 Jps
18906 DataNode
19709 NodeManager
19375 ResourceManager
19119 SecondaryNameNode
```

2. Download and extract Apache Hive

2.1. Download -

a) wgethttps://archive.apache.org/dist/hive/hive-4.0.0/apache-hive-4.0.0-bin.tar.gz

2.2. Extract -

a) tar -xzf apache-hive-4.0.0-bin.tar.gz

b) mv apache-hive-4.0.0-bin hive

3. Configure Environment Varibales

3.1. echo 'export HIVE_HOME=~/Documents/SGSITS/SEM_8/BDA/Assignment_5/hive' >> ~/.bashrc

3.2. echo 'export PATH=$PATH:$HIVE_HOME/bin' >> ~/.bashrc

3.3. source ~/.bashrc

```
SGSITS/SEM_8/BDA/Assignment_5/hive' >> ~/.bashrc
manasvi_ajmera@LAPTOP-61645F5N$ echo 'export PATH=$PATH:$HIVE_HOME/bin' >> ~/.bashrc
manasvi_ajmera@LAPTOP-61645F5N$ source ~/.bashrc
```

4. Prepare HDFS Directories -

4.1. hdfs dfs -mkdir -p /tmp

4.2. hdfs dfs -mkdir -p /user/hive/warehouse

4.3. hdfs dfs -chmod g+w /tmp

4.4. hdfs dfs -chmod g+w /user/hive/warehouse

5. Create hive-site.xml

5.1. nano hive/conf/hive-site.xml

hive-site.xml

hive-site.xml > configuration

```xml
<configuration>
<property>
<name>javax.jdo.option.ConnectionURL</name>
<value>jdbc:derby:/home/manasvi-ajmera/Documents/SGSITS/SEM_8/BDA/Assignment_5/hive/metastore_db;create=true</value>
</property>
<property>
<name>hive.metastore.warehouse.dir</name>
<value>/user/hive/warehouse</value>
</property>
<property>
<name>hive.metastore.warehouse.external.dir</name>
<value>/user/hive/warehouse/external</value>
</property>
<property>
<name>hive.execution.engine</name>
<value>mr</value>
</property>
</configuration>
```

6. Initialize Derby Metastore and Launch

6.1. cd $HIVE_HOME

6.2. bin/schematool -dbType derby -initSchema

```
manasvi_ajmera@LAPTOP-61645F5N$ cd $HIVE_HOME
bin/schematool-dbType derby initSchema
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar: file:/home/manasvi-ajmera/Documents/SGSITS/SEM_8/BDA/Assignment_5/hive/
lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar: file:/home/manasvi-ajmera/Documents/SGSITS/SEM_8/BDA/Assignment_1/hadoo
p-3.4.2/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Initializing the schema to: 4.0.0
Metastore connection URL:
jdbc:derby:;databaseName=metastore_db;create=true
Metastore connection Driver:
org.apache.derby.jdbc. EmbeddedDriver
Metastore connection User:
APP
Starting metastore schema initialization to 4.0.0
Initialization script hive-schema-4.0.0.derby.sql
Initialization script completed
```

7. Start Hive -

7.1. Hive

```
manasvi_ajmera@LAPTOP-61645F5N$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar: file:/home/manasvi_ajmera/Documents/SGSITS/SEM_8/
SLF4J: Found binding in [jar: file:/home/manasvi_ajmera/Documents/SGSITS/SEM_8/
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanatio
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar: file:/home/manasvi_ajmera/Documents/SGSITS/SEM_8/
SLF4J: Found binding in [jar: file:/home/manasvi_ajmera/Documents/SGSITS/SEM_8/
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanatio
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Beeline version 4.0.0 by Apache Hive
beeline> ! connect jdbc:hive2://
Connecting to jdbc:hive2://
Enter username for jdbc:hive2://:
Enter password for jdbc:hive2://:
```

7.2. When the beeline prompt comes - '!connect jdbc:hive2://', as shown in the above image, press Enter for username and password.

7.3. We will that the Hive is connected at the end.

```
Connected to: Apache Hive (version 4.0.0)
Driver: Hive JDBC (version 4.0.0)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://>
```

Hive is connected .